

DM-2026-004

SendCUIEmail Development Team
Software Engineering

DATE: January 21, 2026

TO: Project Record

SUBJECT: Verification Document (VER) Numbering Policy

OFFICE OF PRIMARY RESPONSIBILITY: SendCUIEmail Project

1. **Purpose**

This memorandum establishes the policy for numbering and updating Verification (VER) documents, which provide formal cryptographic compliance verification through code inspection.

2. **Decision**

VER document IDs remain stable across routine version releases. A new VER document ID is only assigned when the cryptographic implementation fundamentally changes.

3. **Policy Details**

Same VER-ID (Dynamic Updates):

The following changes do **not** require a new VER document ID:

- New software version releases (e.g., v0.2.0 → v0.3.0)
- Bug fixes unrelated to cryptographic code
- UI/UX changes
- Documentation updates
- Test infrastructure changes
- Line number shifts due to unrelated code additions

For these changes, the existing VER document is updated with:

- Current version number
- Current git commit hash
- Current commit date
- Updated line number references (if code shifted)

New VER-ID Required:

A new VER document ID (e.g., VER-2026-002) is required when:

- Encryption algorithm changes (e.g., AES-CBC → AES-GCM)
- Key derivation function changes (e.g., PBKDF2 → Argon2)
- Iteration count changes (e.g., 100,000 → 200,000)
- Key size changes (e.g., 256-bit → 512-bit)
- Salt or IV size changes
- Hash algorithm changes (e.g., SHA-256 → SHA-512)
- Any change to cryptographic primitives or parameters

4. Rationale**Verification by Code Inspection:**

VER documents use “verification by code inspection”—a gold standard approach where:

- Each cryptographic requirement is quoted from the authoritative standard
- The corresponding implementation code is shown
- Compliance is demonstrated through direct comparison

Why Stable IDs:

If the cryptographic implementation has not changed, the verification remains valid. The code that was verified in v0.2.0 is the same code in v0.3.0—only the version metadata needs updating.

Creating a new VER-ID for every release would:

- Imply the cryptographic implementation changed (when it didn't)
- Create unnecessary document management overhead
- Dilute the significance of actual cryptographic changes

Why New IDs for Crypto Changes:

When cryptographic primitives change, a new verification is required because:

- Different standards sections may apply
- New compliance requirements must be verified
- The previous verification no longer accurately describes the implementation
- Auditors need clear indication that re-verification occurred

5. Implementation**Automated Updates via test.sh:**

The `test.sh` script automatically updates VER documents before PDF compilation:

- Extracts current version from `CHANGELOG.md`
- Retrieves current git commit hash and date
- Updates VER document title page and metadata using `sed`
- Compiles updated PDF to `.builds/test/`

Updated Fields:

- Title page version (e.g., “SendCUIEmail v0.3.0”)
- Document info table (Version Tested, Git Commit, Commit Date)
- Test verification section commit reference
- Conclusion version/commit reference
- Footer commit reference

Manual Updates Required:

When line numbers shift due to code changes, the following must be manually updated:

- Code excerpt line number references in each requirement section
- This ensures the document accurately points to the verified code

6. Example Scenarios

Change	Action	Rationale
v0.2.0 → v0.3.0 (no crypto changes)	Update VER-2026-001	Same crypto code, same verification
Add FIPS mode detection UI	Update VER-2026-001	UI change, crypto unchanged
Change iterations 100K → 200K	Create VER-2026-002	Crypto parameter changed
Switch AES-CBC to AES-GCM	Create VER-2026-002	Algorithm changed
Add Argon2 as alternative KDF	Create VER-2026-002	New crypto primitive added

7. Document History

Current VER documents:

- **VER-2026-001:** Cryptographic Compliance Verification (NIST SP 800-132, FIPS 197, NIST SP 800-38A, FIPS 140-2)

8. References

- VER-2026-001: Cryptographic Compliance Verification
- NIST SP 800-132: Recommendation for Password-Based Key Derivation
- FIPS 197: Advanced Encryption Standard (AES)
- FIPS 140-2: Security Requirements for Cryptographic Modules
- NIST SP 800-38A: Recommendation for Block Cipher Modes of Operation